

# **AN APPROACH TO STUDY THE COVID-19 STAY-AT-HOME EFFECT ON WATER CONSUMPTION IN LONG ISLAND, NY USING TIME SERIES ANALYSIS OF GROUNDWATER LEVELS**

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# Introduction

- Bhat et al., 2021 (India & other European nations); Gualtieri et al.(Italy), 2020; Ming, 2020 (China)
  - COVID-19 & environmental factors: air and water (quality and not the quantity).
- Kalbusch, 2021 (Joinville, Southern Brazil)
  - Commercial, industrial, and public water consumption dropped
  - Residential rates rose
  - Overall water consumption dropped
- Sayeed et al., 2021 (Bangladesh)
  - 1,134/ 1,980 participants (57.27%) left the water running to scrub their hands.
  - Wasted approximately 12.79x more water per day than they had before frequent hand washing protocols.
- For this study
  - Results were predicted to show a decrease in groundwater levels as excessive water was consumed over 2020.
  - Data: LI USGS Groundwater stations, 2000- 2021
  - Analysis: normalize (kza) function in R studio

# Methodology

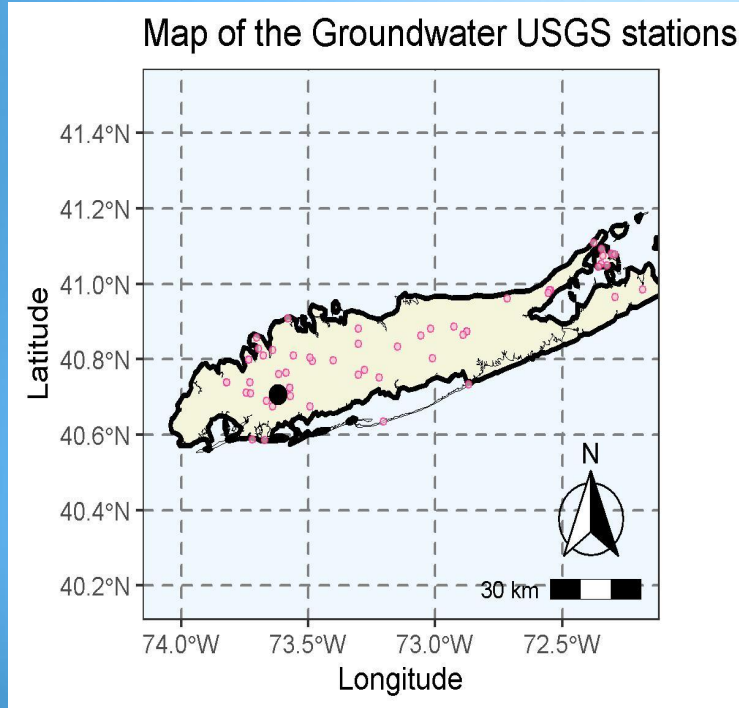
- Data used: USGS GWL 2000-2021
- R programming language with functions of moving average and filtering
- Lollipop diagram is created using ggplot2 package to present groundwater levels data from all the USGS stations
- `kz()` function is a moving average function used to filter high frequencies (low periodicities) to reveal long-term trends graphs.
- The `periodogram()` function by TSA (time series analysis) is used to perform spectral analysis and it determines the appropriate window size for moving average: (30,5) (90,5) (365,5)
- The moving average function in R Studio is designed to average a specific window of data from the overall data to clarify trend(s) that occur within those frames and eliminate irregular frequencies/noise over the entire time series dataset

# Brief Description of Kolmogorov-Zurbenko (KZ) low-pass filter

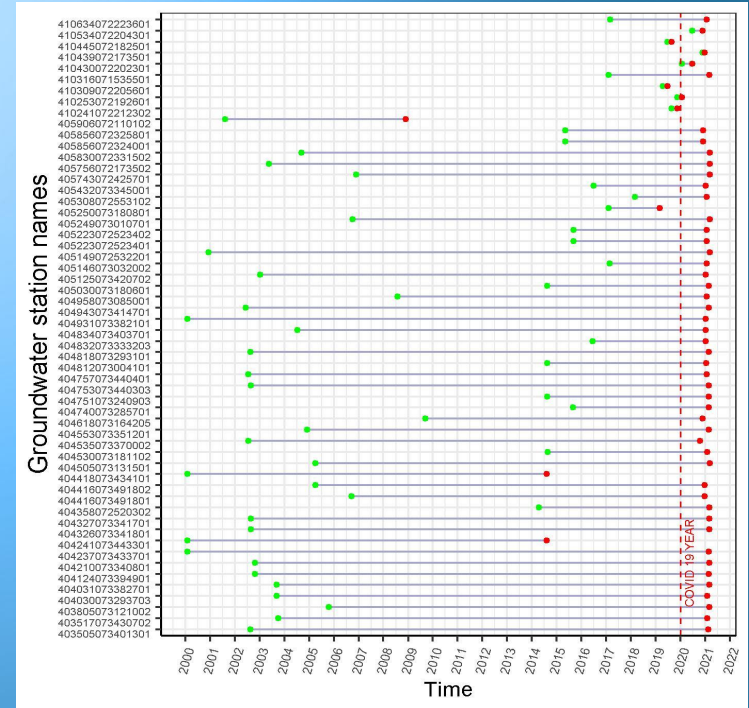
$$Y_t = \frac{1}{m} \sum_{j=-k}^k (X_{t+j}) \quad (1)$$

$$Y_t = \text{KZ}_{m,p}(X_t) \quad (2)$$

# Results

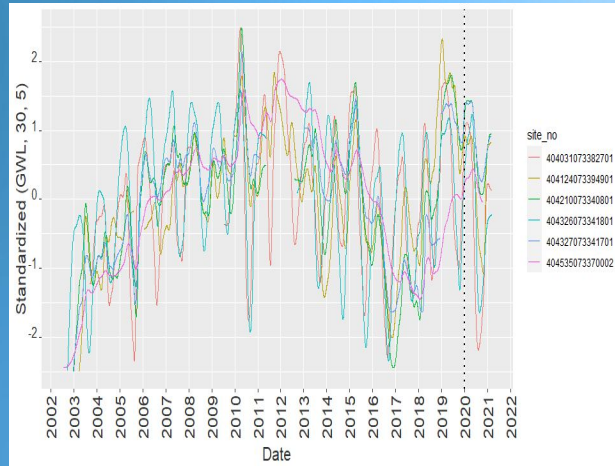


**Map of Groundwater stations on Long Island**

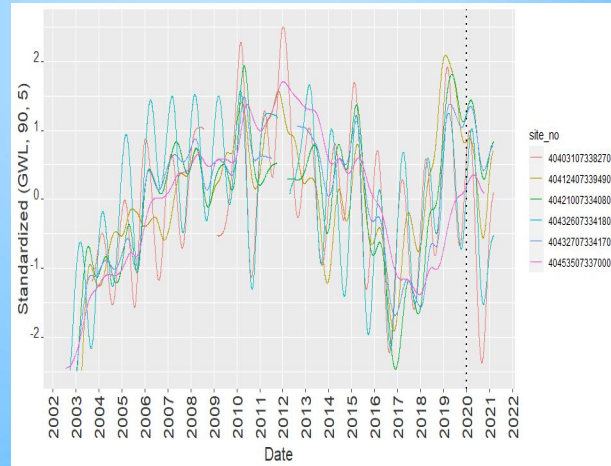


**Lollipop Diagram of all Long Island USGS GWL**

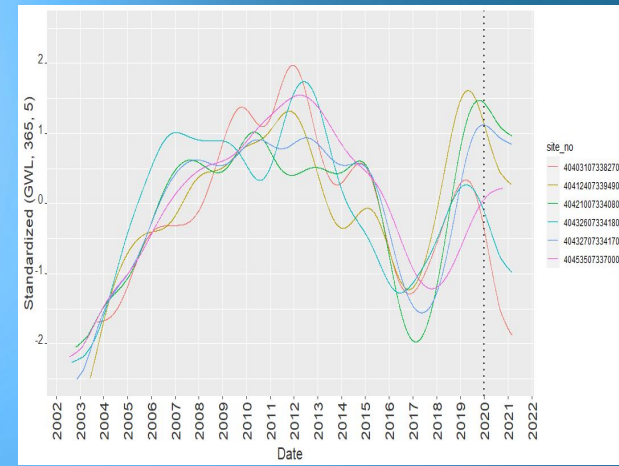
# Results



**30-day MA with  $kz(30,5)$  after  
`normalize()` in R using USGS  
groundwater levels for stations near  
Hempstead, NY**

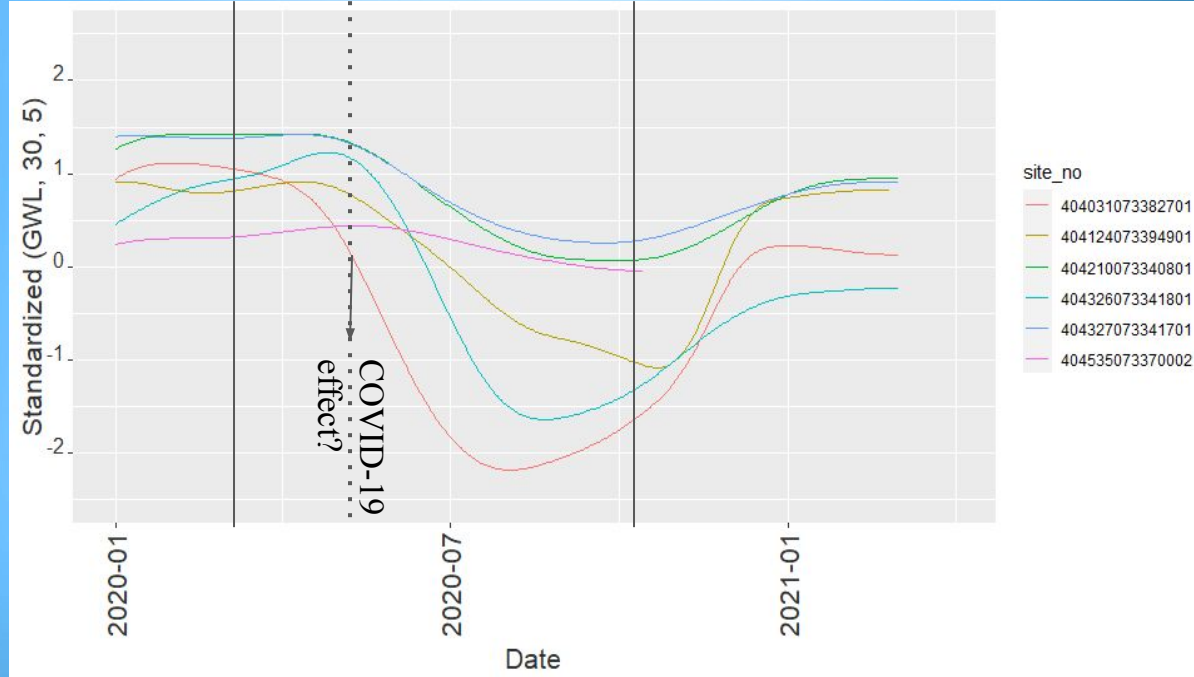


**90-day MA with  $kz(90,5)$  after  
`normalize()` in R using USGS  
groundwater levels for stations near  
Hempstead, NY**



**365-day MA with  $kz(365,5)$  after  
`normalize()` in R using USGS  
groundwater levels for stations near  
Hempstead, NY**

# Results



**30-day MA with  $kz(30,5)$  after `normalize()` in R using  
USGS groundwater levels for stations near Hempstead,  
NY with a focus in 2020**

# Discussion

- With frequent rainfall and snowstorms, Long Island experiences a surplus in groundwater budget, however groundwater overconsumption could result in an influx of salt water contamination into the groundwater system and lead to higher withdrawals if groundwater supply is not well managed during crises such as a drought or a pandemic (Fetter, 2014).
- Groundwater data trends of both short and long term assessments indicate a significant drop in groundwater levels contrasting with the events of the COVID-19 pandemic, with residential and industrial regions as the source for groundwater withdrawal.
- The COVID-19 pandemic may pose as a statistical outlier for future research into groundwater data, in and around Long Island, as shown by the downward trend after a 15-year upward trend was expected. It will perpetually be displayed as an anomaly in future analyses (as seen by moving average graph).

# Conclusion

- The groundwater levels shown in the 30-, 90-, and 365- moving averages may imply an overall consensus of water consumption in Hempstead, NY increasing during the COVID-19 lockdown (graphs with MAs).
- Over a 30-day moving average, a downward trend starts appearing at the beginning of COVID-19 lockdown and continues down this trend until the end of 2020, when lockdown restrictions were lifted.
- The hypothesis that water usage would increase drastically throughout the lockdown period from the COVID-19 pandemic is supported and suggests that groundwater was more heavily consumed over the year 2020 and may continue to be a prominent irregularity for the future groundwater trend.

## Acknowledgements

- Hofstra
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## Credit Authorship

Buchbinder, B: data curation, R coding, writing - Introduction, Methodology, Conclusion, editing; Phan, E.: R coding, data formatting & formal analysis, writing - Introduction, Methodology, Conclusion, Discussion, editing; Renner, F.: writing - Abstract, Introduction, Conclusion, Discussion, editing; Liebowitz, A.: reference citing, editing; Achek, M.: writing - Conclusion, Discussion, editing; Marsellos, A.E.: Supervision, Conceptualization, R coding, editing; Tsakiri, K.G.: R coding, writing - Methodology

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